

Packet Tracer – Static Routing with Multiple Routers

Objectives

Part 1: Build a Multi-Router Network Topology

Part 2: Configure IP Addressing

Part 3: Configure Static Routes

Part 4: Verify Routing Tables

Part 5: Test End-to-End Connectivity

Background / Scenario

Routers can only communicate with networks that are directly connected to them unless they learn about remote networks through routing. One way to achieve this is through **Static Routing**, where administrators manually configure routes.

In this lab, you will connect three routers together, configure IP addressing on all interfaces, create static routes, and verify communication between devices on different networks.

This lab focuses on:

- Configuring router interfaces
- Creating static routes
- Understanding routing tables
- Testing end-to-end connectivity
- Verifying packet forwarding between routers

You are developing a foundational routing skill that is essential before learning dynamic routing protocols such as RIP, OSPF, and EIGRP.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 3 Routers (1941 or 2911)
- 2 Switches (2960)
- 2 PCs

c. Rename the devices:

- Router0 → **R1**
 - Router1 → **R2**
 - Router2 → **R3**
 - Switch0 → **S1**
 - Switch1 → **S2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Device Interface		Device Interface	
PC-A	FastEthernet0	S1	Fa0/1
S1	Gi0/1	R1	Gi0/0
R1	Gi0/1	R2	Gi0/0
R2	Gi0/1	R3	Gi0/0
R3	Gi0/1	S2	Gi0/1
S2	Fa0/1	PC-B	FastEthernet0

Wait until the links turn green.

Network Addressing Plan

Device Interface IP Address

PC-A	Fa0	192.168.1.10/24
R1	Gi0/0	192.168.1.1/24
R1	Gi0/1	10.0.12.1/30
R2	Gi0/0	10.0.12.2/30
R2	Gi0/1	10.0.23.1/30
R3	Gi0/0	10.0.23.2/30
R3	Gi0/1	192.168.3.1/24
PC-B	Fa0	192.168.3.10/24

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Go to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Step 2: Configure PC-B

Enter:

- IP Address: 192.168.3.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1

Step 3: Configure R1 Interfaces

Open the CLI and enter:

enable

configure terminal

interface gigabitethernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitethernet 0/1

ip address 10.0.12.1 255.255.255.252

no shutdown

exit

Step 4: Configure R2 Interfaces

enable

configure terminal

interface gigabitethernet 0/0

ip address 10.0.12.2 255.255.255.252

no shutdown

exit

interface gigabitethernet 0/1

ip address 10.0.23.1 255.255.255.252

no shutdown

exit

Step 5: Configure R3 Interfaces

enable

configure terminal

```
interface gigabitethernet 0/0
ip address 10.0.23.2 255.255.255.252
no shutdown
exit
```

```
interface gigabitethernet 0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

Part 3: Verify Directly Connected Networks

Step 1: Test Neighbor Connectivity

From R1:

```
ping 10.0.12.2
```

From R2:

```
ping 10.0.23.2
```

Both pings should succeed.

Step 2: View Routing Tables

On each router, type:

```
show ip route
```

Notice that only directly connected networks appear.

Example on R1:

```
C 192.168.1.0/24 is directly connected
```

```
C 10.0.12.0/30 is directly connected
```

R1 does not yet know about the 192.168.3.0 network.

Part 4: Configure Static Routes (Enter Global Configuration Mode)

Step 1: Configure Static Route on R1

To reach the network behind R3, enter:

```
ip route 192.168.3.0 255.255.255.0 10.0.12.2
```

Step 2: Configure Static Routes on R2

```
ip route 192.168.1.0 255.255.255.0 10.0.12.1
```

```
ip route 192.168.3.0 255.255.255.0 10.0.23.2
```

Step 3: Configure Static Route on R3

To reach the network behind R1:

```
ip route 192.168.1.0 255.255.255.0 10.0.23.1
```

Why Are Static Routes Needed?

Each router only knows:

- Its directly connected networks
- Networks manually configured by the administrator

Static routes tell the router:

"If you need to reach this network, send the packet to this next-hop router."

Part 5: Verify the Routing Table

Step 1: Check R1

Type:

```
show ip route
```

Expected output:

```
S 192.168.3.0/24 [1/0] via 10.0.12.2
```

The "S" indicates a Static Route.

Step 2: Check R3

Type:

show ip route

Expected output:

S 192.168.1.0/24 [1/0] via 10.0.23.1

Part 6: Test End-to-End Connectivity

Step 1: Ping from PC-A to PC-B

On PC-A:

ping 192.168.3.10

Expected Result:

Reply from 192.168.3.10

You should receive four successful replies.

Step 2: Ping from PC-B to PC-A

On PC-B:

ping 192.168.1.10

You should again receive successful replies.

Step 3: Trace the Route

On PC-A:

tracert 192.168.3.10

Expected path:

192.168.1.1

10.0.23.2

192.168.3.10

The packet travels through multiple routers before reaching its destination.

Understanding the Packet Path

PC-A

|

S1

|

R1

|

R2

|

R3

|

S2

|

PC-B

The routers use static routes to determine where packets should be forwarded.

Troubleshooting Tips

If the ping fails:

✓ Verify all interfaces are enabled using:

show ip interface brief

✓ Confirm all IP addresses are correct.

✓ Verify static routes using:

show ip route

✓ Check that default gateways are configured on both PCs.

✓ Ensure all links are green.

✓ Test router-to-router connectivity before testing PC-to-PC connectivity.

What You Learned

- How routers forward packets between networks.
 - How to configure router interfaces.
 - How to create static routes.
 - How to verify routes using show ip route.
 - How to troubleshoot routing issues.
 - How packets travel across multiple routers.
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Challenge Extensions (Optional)

Try modifying the lab:

1. Add a fourth router and another remote network.
2. Replace static routing with RIP.
3. Replace static routing with OSPF.
4. Remove one static route and observe the effect on connectivity.
5. Use Simulation Mode to watch packets move through all three routers.